

Tutorial 4

1. Specific Excess Power (SEP) represents:
 - (a) an aircraft's flight envelope;
 - (b) an aircraft's specific energy level throughout the aircraft's envelope;
 - (c) the effect of altitude and airspeed on an aircraft's drag curve.
 - ☒ (d) the level of excess thrust available throughout the aircraft's envelope;
2. Specific energy or energy height represents
 - ☒ (a) the aircraft's total energy per unit weight;
 - (b) the rate of change of an aircraft's energy with height;
 - (c) the aircraft's total energy per unit mass;
 - ☒ (d) the aircraft's kinetic energy per unit mass at any given height;
3. The $P_S = 0$ contour indicates:
 - (a) maximum flight speed and minimum flight speed
 - ☒ (b) maximum flight speed and absolute ceiling
 - (c) maximum flight speed and service ceiling
 - (d) maximum flight speed, minimum flight speed and absolute ceiling
4. Which of the following do SEP contours take into consideration:
 - ☒ (a) all wing design characteristics;
 - (b) pilot limits;
 - ☒ (c) thrust lapse;
 - (d) structural design limits.
5. Which of the following do SEP curves not depend on?
 - (a) Aircraft weight.
 - ☒ (b) Pilot technique.
 - (c) Load factor.
 - (d) Throttle setting.

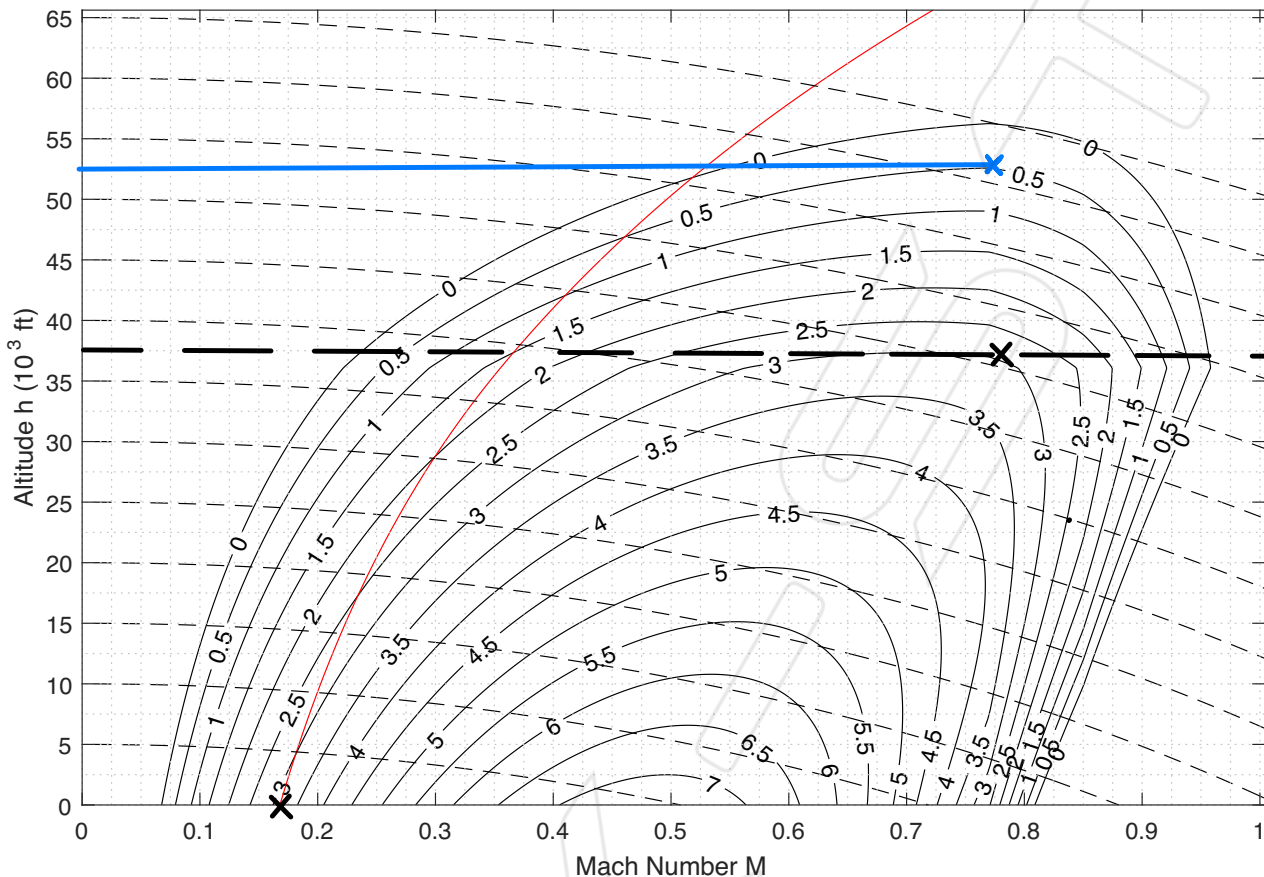


Figure 1: Plot of Specific Excess Power contours P_S (1000 ft/min) for a jet airliner, with Energy Heights (- -) and Stall curve (in red) superimposed

6. Which of the following pieces of information cannot be extracted from the SEP plot given in figure 1?
 - (a) The stall speed is approximately 111 kts EAS. $0.17 \times 340.7 \text{ m/s}$
 - (b) The maximum rate of climb possible at 37,500 ft is approximately 3,000 ft/min.
 - ☒ (c) The maximum aircraft (L/D) is approximately 17.7. $\frac{Y}{L/D} = \frac{VT_x}{W}$ when $P_S = 0$
 - (d) The best cruise speed is approximately 200 kts EAS.
7. The service ceiling of the aircraft whose SEP plot is given in figure 1 will be:
 - (a) 56,000 ft;
 - (b) 37,000 ft;
 - ☒ (c) 52,500 ft;
 - (d) 65,000 ft.
8. Which velocity “schedule” should a pilot follow to climb in the minimum time possible?
 - ☒ (a) The velocities where SEP contours are tangent to energy height lines for each altitude.
 - (b) The velocities where SEP contours are normal to energy height lines for each altitude.
 - (c) The velocities where rate of climb is maximised for each altitude.
 - (d) The maximum velocity achievable at each altitude.

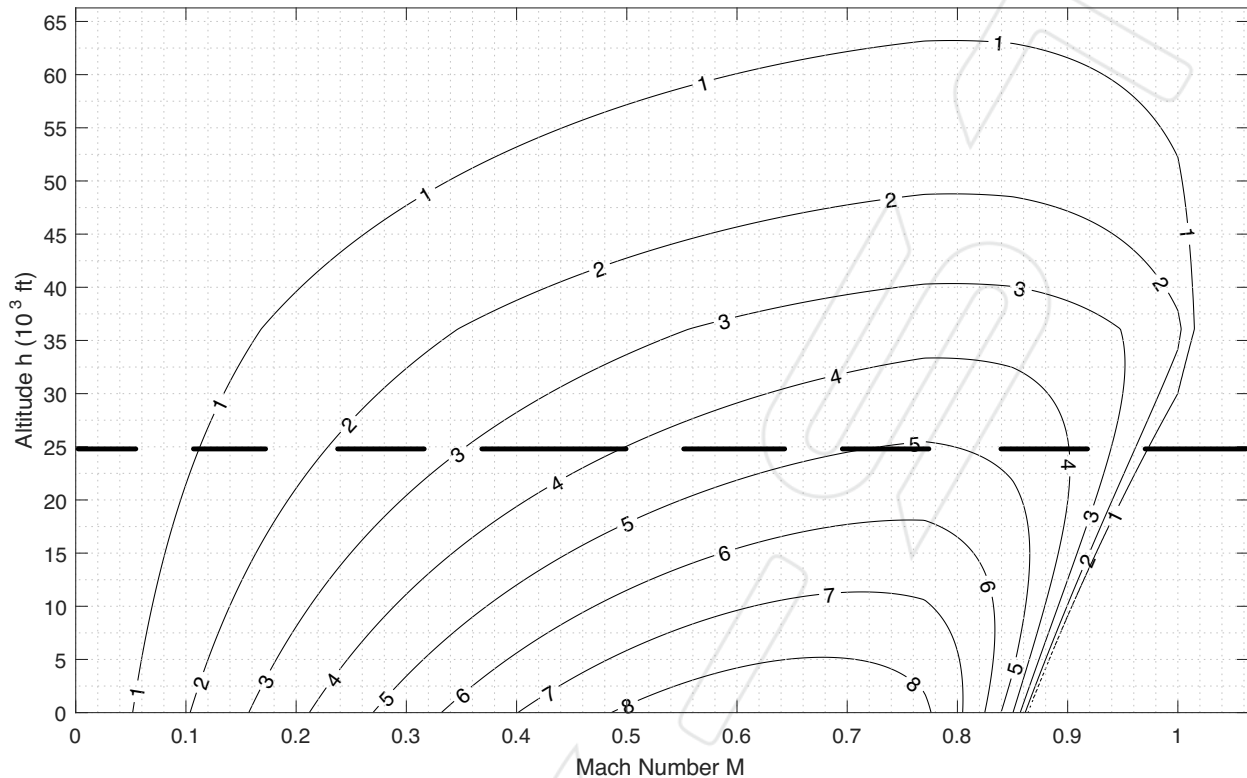


Figure 2: $P_S = 0$ contours at varying load factor, n , for a low-powered fighter.

9. The fighter whose SEP contours are given in Figure 2 is engaged in aerial combat. What would the radius of the tightest level turn achievable by the pilot with a 70° bank angle at an altitude of 25 000 ft?

$$n = \frac{L}{W} = \frac{L}{L \cos \phi} = \frac{1}{\cos \phi} = 2.92$$

$$R = \frac{V_{mo}^2}{g} \frac{\bar{V}^*}{\sqrt{n^2 - 1}}$$

(a) 460 m **If $M_{stall} < 0.3$*

(b) 3,150 m

(c) 4,533 m

☒ (d) There is insufficient info to approximate.

10. What is the maximum load factor that the pilot of the fighter whose SEP contours are given in Figure 2 can achieve when flying at an altitude of 25 000 ft?

(a) There is insufficient info to approximate.

(b) Approximately 1 g;

(c) Approximately 3 g;

☒ (d) Approximately 5 g;

SI Units →		The Standard Atmosphere							A.101/102
Alt. Feet	Alt. m.	T/K	a	ρ	σ	$\sqrt{\sigma}$	p/p0	p (Pa)	μ
0	0	288.2	340.7	1.225	1.0000	1.0000	1.0000	101325	1.795E-05
1640	500	284.9	338.8	1.167	0.9529	0.9762	0.9421	95463	1.779E-05
3281	1000	281.7	336.9	1.112	0.9075	0.9526	0.8870	89878	1.763E-05
4921	1500	278.4	334.9	1.058	0.8638	0.9294	0.8346	84561	1.747E-05
6562	2000	275.2	332.9	1.007	0.8217	0.9065	0.7846	79502	1.731E-05
8202	2500	271.9	331.0	0.957	0.7812	0.8838	0.7371	74690	1.715E-05
9843	3000	268.7	329.0	0.909	0.7422	0.8615	0.6920	70117	1.699E-05
11483	3500	265.4	327.0	0.863	0.7048	0.8395	0.6491	65774	1.683E-05
13123	4000	262.2	325.0	0.819	0.6688	0.8178	0.6084	61651	1.666E-05
14764	4500	258.9	323.0	0.777	0.6342	0.7964	0.5698	57739	1.650E-05
16404	5000	255.7	320.9	0.736	0.6010	0.7753	0.5332	54031	1.633E-05
18045	5500	252.4	318.9	0.697	0.5692	0.7545	0.4986	50519	1.617E-05
19685	6000	249.2	316.8	0.660	0.5387	0.7339	0.4658	47193	1.600E-05
21325	6500	245.9	314.7	0.624	0.5094	0.7137	0.4347	44047	1.583E-05
22966	7000	242.7	312.7	0.590	0.4814	0.6938	0.4054	41073	1.566E-05
24606	7500	239.4	310.6	0.557	0.4545	0.6742	0.3776	38264	1.549E-05
26247	8000	236.2	308.4	0.525	0.4289	0.6549	0.3515	35612	1.532E-05
27887	8500	232.9	306.3	0.495	0.4043	0.6358	0.3268	33112	1.515E-05
29528	9000	229.7	304.2	0.467	0.3808	0.6171	0.3035	30755	1.498E-05
31168	9500	226.4	302.0	0.439	0.3584	0.5987	0.2816	28536	1.480E-05
32808	10000	223.2	299.8	0.413	0.3371	0.5806	0.2610	26448	1.463E-05
34449	10500	219.9	297.6	0.388	0.3167	0.5627	0.2417	24486	1.445E-05
36089	11000	216.66	295.4	0.364	0.2972	0.5452	0.2235	22644	1.427E-05
37730	11500	216.7	295.5	0.336	0.2747	0.5241	0.2066	20931	1.428E-05
39370	12000	216.7	295.5	0.311	0.2539	0.5039	0.1909	19344	1.428E-05
41010	12500	216.7	295.5	0.287	0.2346	0.4844	0.1764	17877	1.428E-05
42651	13000	216.7	295.5	0.266	0.2168	0.4657	0.1631	16522	1.428E-05
44291	13500	216.7	295.5	0.245	0.2004	0.4477	0.1507	15270	1.428E-05
45932	14000	216.7	295.5	0.227	0.1852	0.4303	0.1393	14112	1.428E-05
47572	14500	216.7	295.5	0.210	0.1712	0.4137	0.1287	13042	1.428E-05
49213	15000	216.7	295.5	0.194	0.1582	0.3977	0.1190	12053	1.428E-05
50853	15500	216.7	295.5	0.179	0.1462	0.3824	0.1099	11139	1.428E-05
52493	16000	216.7	295.5	0.166	0.1351	0.3676	0.1016	10295	1.428E-05
54134	16500	216.7	295.5	0.153	0.1249	0.3534	0.0939	9514	1.428E-05
55774	17000	216.7	295.5	0.141	0.1154	0.3397	0.0868	8793	1.428E-05
57415	17500	216.7	295.5	0.131	0.1067	0.3266	0.0802	8127	1.428E-05
59055	18000	216.7	295.5	0.121	0.0986	0.3140	0.0741	7510	1.428E-05
60696	18500	216.7	295.5	0.112	0.0911	0.3018	0.0685	6941	1.428E-05
62336	19000	216.7	295.5	0.103	0.0842	0.2901	0.0633	6415	1.428E-05
63976	19500	216.7	295.5	0.095	0.0778	0.2789	0.0585	5928	1.428E-05
65617	20000	216.7	295.5	0.088	0.0719	0.2682	0.0541	5479	1.428E-05

NB: Conversion factors - 1 knot = 1.689 ft/s = 0.5148 m/s

Long Question

Figure 3 shows the specific excess power (P_S) plot obtained when flight testing a jet trainer at a wing loading $W/S = 4937 \text{ N/m}^2$. The aircraft has the following characteristics:

- wing reference area $S = 16.7 \text{ m}^2$,
- wing aspect ratio $R = 7$,
- loading efficiency $k = 1.25$,
- specific fuel consumption $c = 0.45 \text{ 1/hour}$
- maximum sea-level thrust to weight ratio $T_0/W = 0.352$.

The aircraft's thrust may be assumed to be independent of Mach number and dependent on altitude only. In the troposphere you may use $T/T_0 = \sigma^{0.7}$ and $T/T_0 = 1.439\sigma$ in the stratosphere.

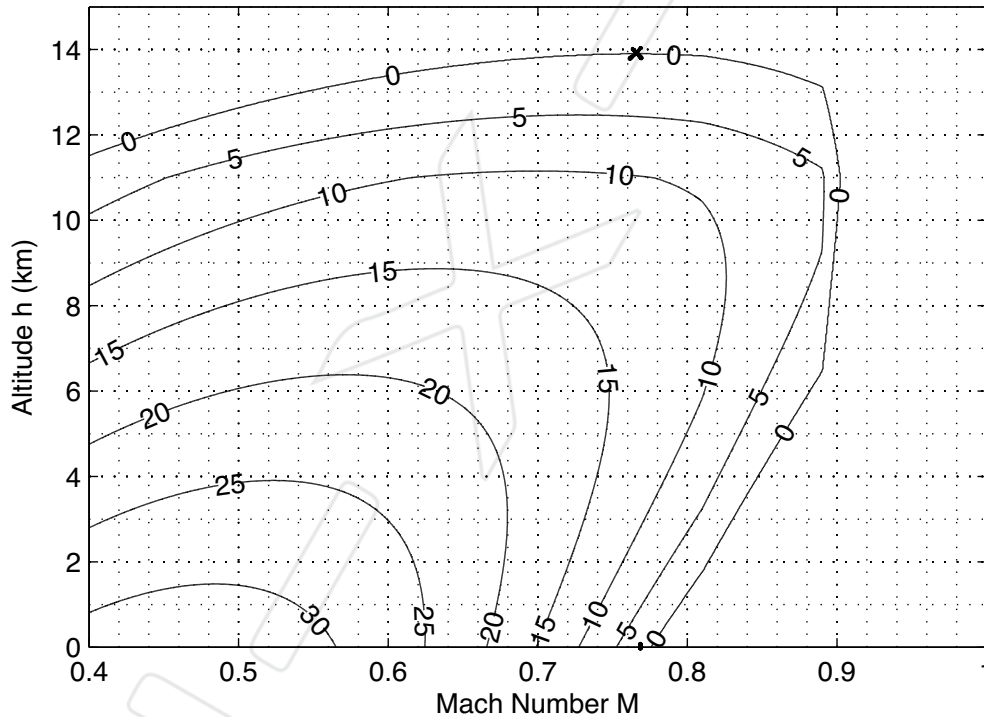


Figure 3: Variation of specific excess power (in m/s) with altitude and Mach number

Based on Figure 3:

1. approximately calculate the aircraft's minimum drag Equivalent Airspeed V_{imD} ; [20%]
2. approximately calculate the aircraft zero-lift drag coefficient C_{D0} ; [35%]
3. what is the maximum endurance of this aircraft if 1400 kg of fuel is available? State any assumptions made. [25%]
4. what is the maximum climb gradient the aircraft can achieve at sea-level at the end of its loiter? [20%]

$$1. M_{mo} = 0.76$$

$$V_{mo} (13.8 \times 10^3 \text{ m} \times 14000 \text{ m}) = 0.76 \times 295.5 = 224.58 \text{ (m/s)}$$

$$\therefore V_{ind} = V_{mo} \sqrt{6} = 0.4303 V_{mo} = 96.6 \text{ (m/s)}$$

2. At absolute ceiling,

$$L/D = (L/D)_{max} = \frac{1}{2} \sqrt{\frac{\pi R}{k C_{D0}}}$$

$$\text{When } P_s = 0, V \left(\frac{T_x}{W} - \frac{1}{L/D} \right) = 0$$

$$\therefore (L/D)_{max} = \frac{W}{T} = \frac{W}{T_0 \cdot 1.4396} \text{ at } 13800 \text{ m} = 10.660$$

$$\therefore C_{D0} = \frac{\pi R}{[2(L/D)_{max}]^2 k} = 0.0387$$

$$3. E = -\frac{(L/D)_{max}}{C} \left[\frac{2}{\bar{V}^2 + \bar{V}^{-2}} \right] \ln \left(\frac{W_f}{W_i} \right)$$

$$\text{While } \bar{V} = 1, E_{max} = -\frac{(L/D)_{max}}{C} \ln \left(\frac{W_f}{W_i} \right) = 4.316 \text{ h}$$

$$4. P_s = V \left(\frac{T_x}{W} - \frac{1}{L/D} \right) \text{ while } V \text{ is constant}$$

$$\sin \delta = \frac{T-D}{W_f} = \frac{T_0}{W_f} - \frac{D}{W_f} = \frac{T_0}{W_f} - \frac{1}{(L/D)_{max}} \quad (L \approx W \text{ for small } \delta)$$

$$\therefore \delta_{max} = 19.2^\circ$$